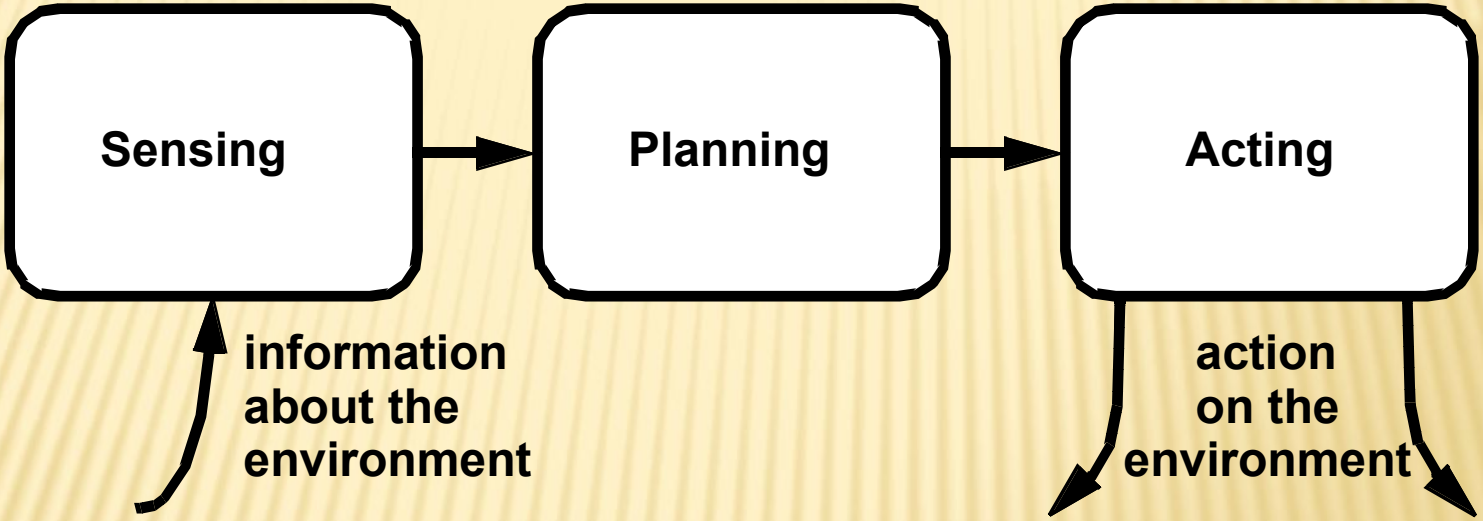


SENSORS

WHAT MAKES A MACHINE A ROBOT?



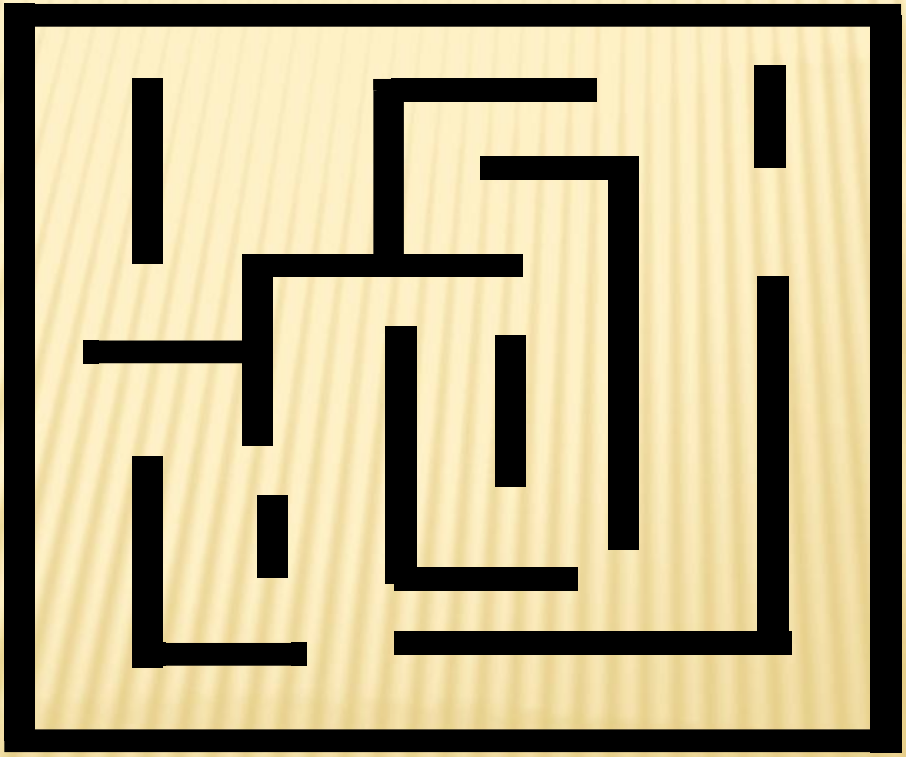






WHY DO ROBOTS NEED SENSORS?

Where am I?



localization

WHY DO ROBOTS NEED SENSORS?

Will I hit anything?





obstacle detection

WHY DO ROBOTS NEED SENSORS?

Sensing for specific tasks

Where is the cropline?





Autonomous harvesting

WHY DO ROBOTS NEED SENSORS?

Sensing for specific tasks

Where is the face?



Face detection & tracking

SENSORS & DETECTORS

- A sensor: is a device that measures a physical quantity and converts it into an electrical signal which can be read by an instrument
- A sensor job :is to convert a parameter in one form and report it in another form of energy
- For example a pressure sensor converts pressure to electricity for display at a remote gauge
- A detector: it is a device that receives a signal as heat or pressure or light or motion and responds to it in a distinctive manner
- Usually a detector consists of a sensor to measure the physical quantity and a decision circuit to respond appropriately.
- Smoke detector uses a smoke sensor to measure the smoke level, then it will produce and alarm if the smoke level is high

SENSORS

- There are a vast number of different sensors being used in robotics, applying different *measurement techniques*, and using different *interfaces* to a controller.
- This, unfortunately, makes sensors a difficult subject to cover.
- We will, however, select a number of typical sensor systems and discuss their details in hardware and software.
- The scope of this chapter is more on interfacing sensors to controllers than on understanding the internal construction of sensors themselves.
- What is important is to find the right sensor for a particular application
  - ✓ Measurement technique
  - ✓ Size
  - ✓ Weight
  - ✓ Operating temperature range
  - ✓ Power consumption
  - ✓ Price range

SENSORS DATA TRANSFER

- Data transfer from the sensor to the CPU can be either
  1. CPU-initiated (*polling*) or
  2. Sensor-initiated (via *interrupt*).
- In CPU-initiated, the CPU has to keep checking whether the sensor is ready by reading a status line in a loop.
- This is much more time consuming than the sensor-initiated data transfer
- In Sensor initiated data transfer requires the availability of an interrupt line.
- The sensor signals via an interrupt that data is ready, and the CPU can react immediately to this request

SENSOR CATEGORIES

- From an engineer's point of view, it makes sense to classify sensors according to their output signals.
- This will be important for interfacing them to an embedded system.

Sensor Output	Sample Application
Binary signal (0 or 1)	Tactile sensor
Analog signal (e.g. 0..5V)	Inclinometer
Timing signal (e.g. PWM)	Gyroscope
Serial link (RS232 or USB)	GPS module
Parallel link	Digital camera

SENSOR CATEGORIES

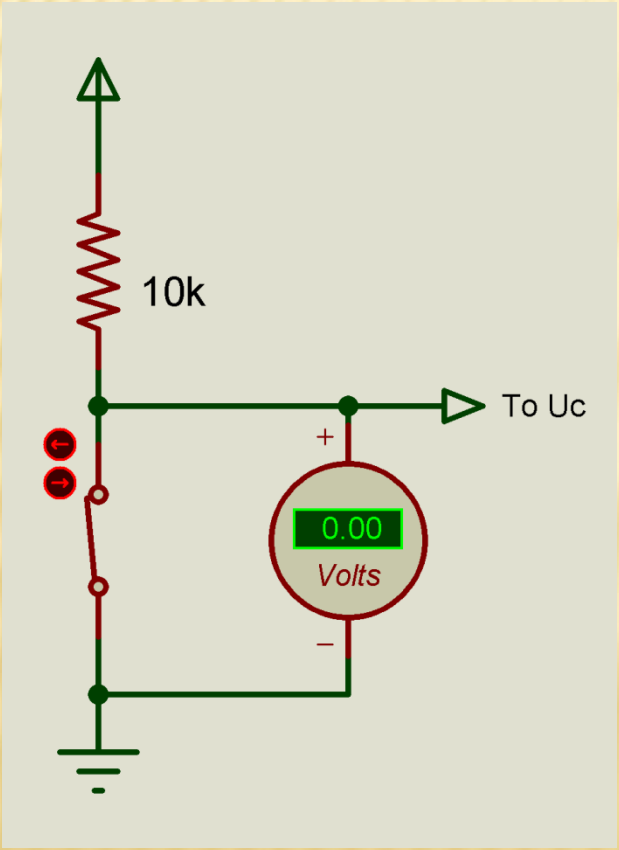
- A different classification is required when looking at the application side
- From a robot's point of view, it is more important to distinguish:
  - Local or on-board sensors (*sensors mounted on the robot*)
  - Global sensors (sensors mounted outside the robot in its environment and transmitting sensor data back to the robot)
- For mobile robot systems it is also important to distinguish:
  - Internal sensors (sensors monitoring the robot's internal state)
  - External sensors (sensors monitoring the robot's environment)
- A further distinction is between:
  - Passive sensors
    Sensors that monitor the environment without disturbing it, for example digital camera, gyroscope
  - Active sensors
    Sensors that stimulate the environment for their measurement, for example sonar sensor, laser scanner, infrared sensor

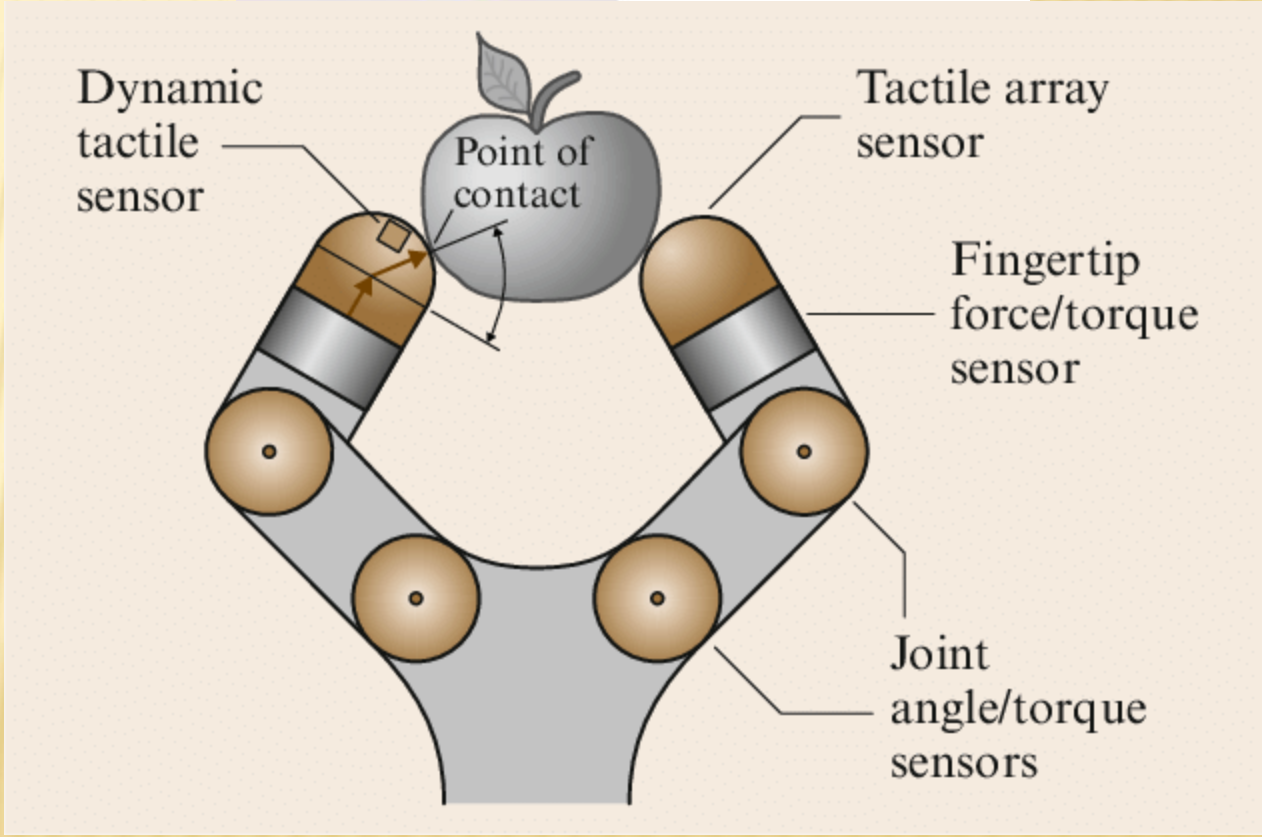
SENSOR CLASSIFICATION

	Local	Global
Internal	**Passive** battery sensor, chip-temperature sensor, shaft encoders, accelerometer, gyroscope, inclinometer, compass **Active** –	**Passive** – **Active** –
External	**Passive** on-board camera **Active** sonar sensor, infrared distance sensor, laser scanner	**Passive** overhead camera, satellite GPS **Active** sonar (or other) global positioning system

BINARY SENSOR

- Binary sensors are the simplest type of sensors.
- They only return a single bit of information, either 0 or 1.
- A typical example is a tactile sensor on a robot.



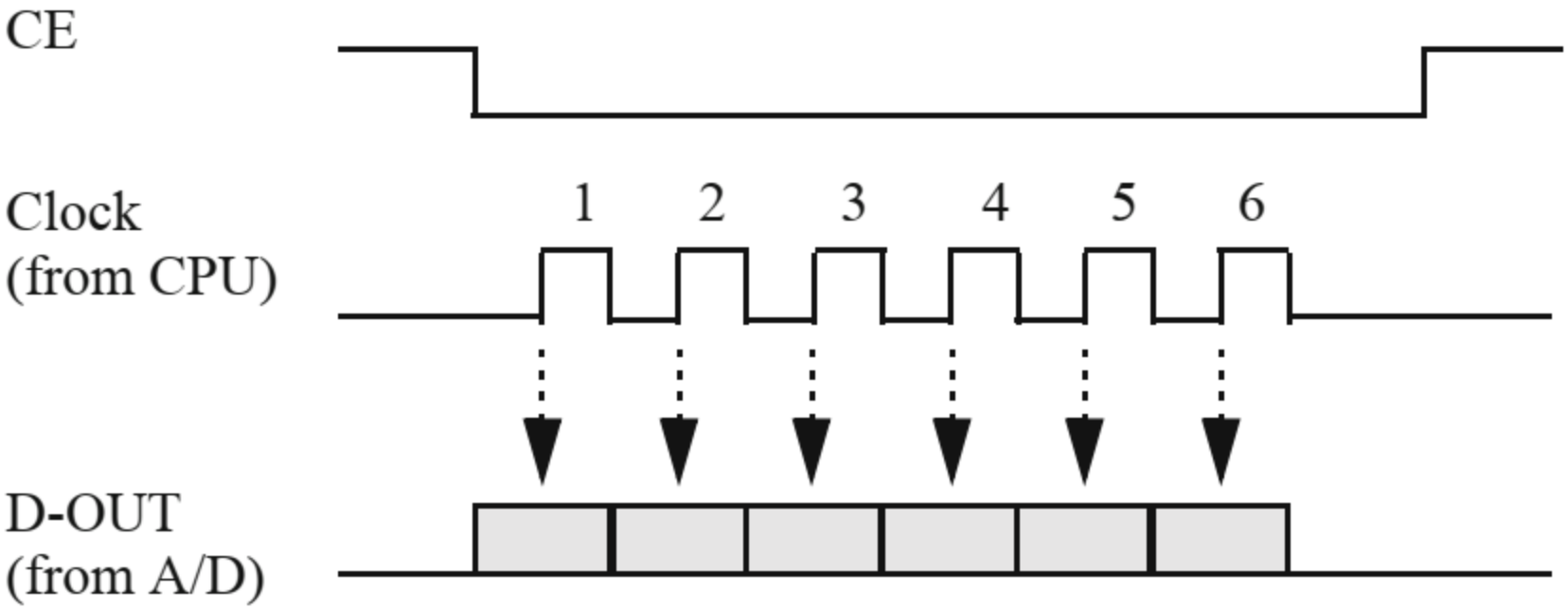


ANALOG VERSUS DIGITAL SENSORS

- A number of sensors produce analog output signals rather than digital signals
- This means an A/D converter is required to connect such a sensor to a microcontroller
- Typical examples:
  - Microphone
  - Analog infrared distance sensor
  - Analog compass
  - Barometer sensor
- Digital sensors on the other hand are usually more complex than analog sensors and often also more accurate
- The output signal of digital sensors can have different forms.
  - It can be a parallel interface (for example 8 or 16 digital output lines) or
  - A serial interface
    (for example following the RS232 standard) or a “synchronous serial” interface.

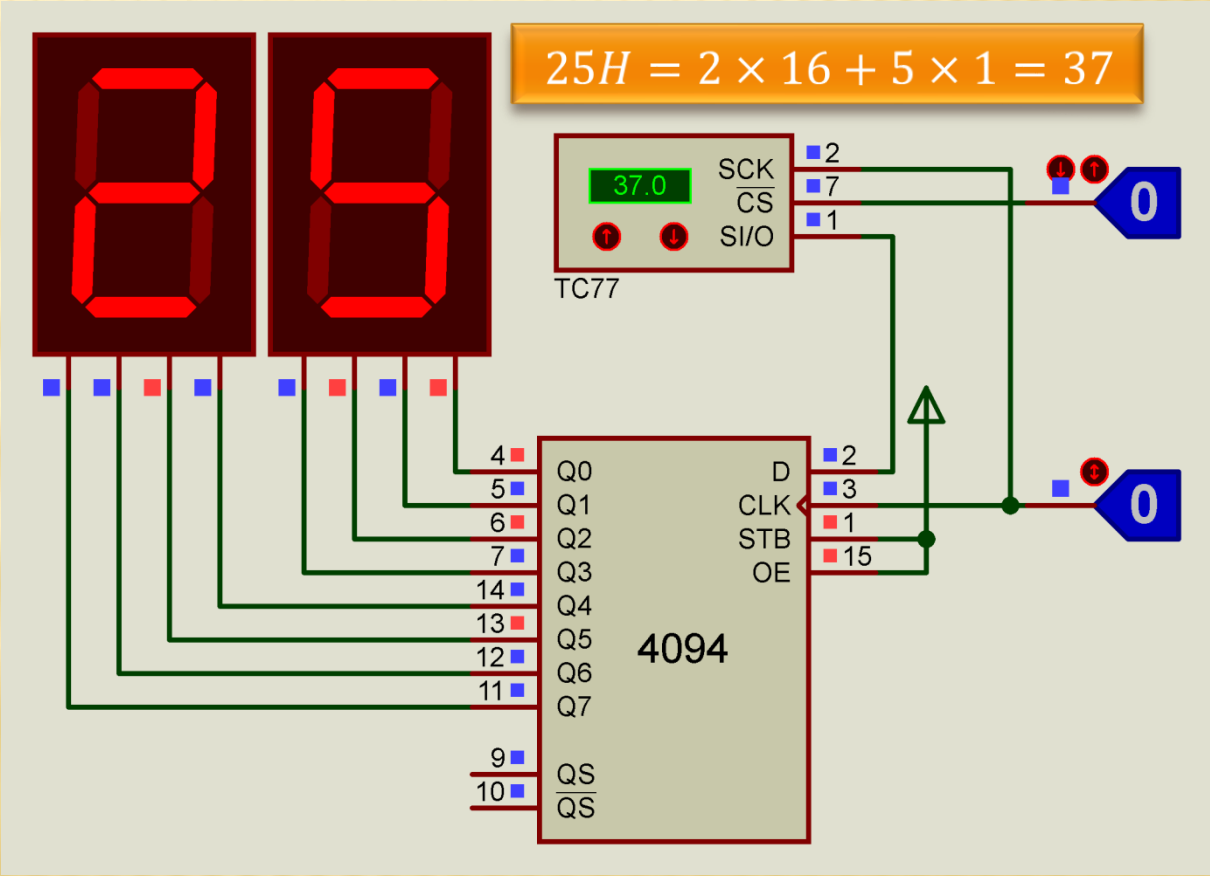
ANALOG VERSUS DIGITAL SENSORS

- The expression “synchronous serial” means that the converted data value is read bit by bit from the sensor.
- After setting the chip-enable line for the sensor, the CPU sends pulses via the serial clock line and at the same time reads 1 bit of information from the sensor’s D output line (for example on each rising edge
- The following is an example of a sensor with a 6bit wide output word



TC72 TEMPERATURE SENSOR EXAMPLE

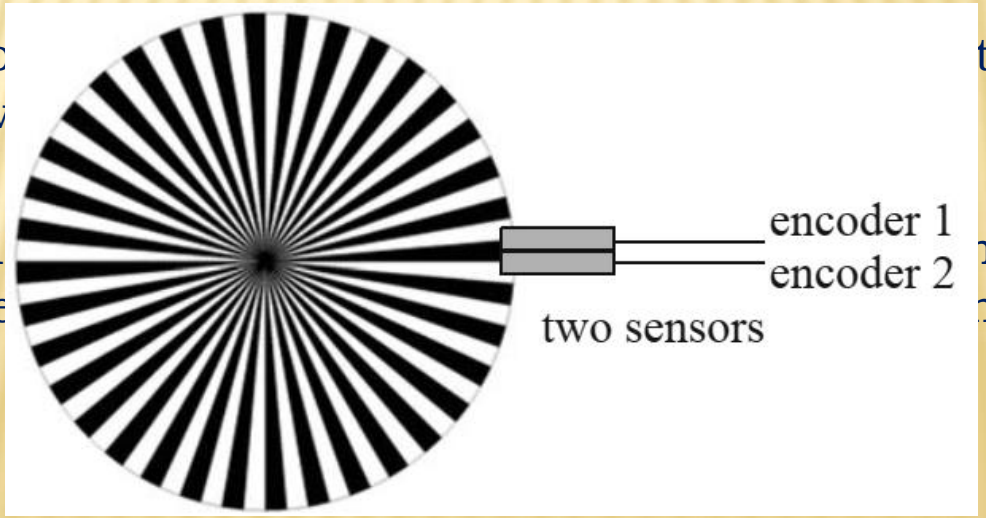
- The TC72 is a temperature sensor with synchronous serial digital output interface.
- A simple shift register can be used to convert the serial data to 8-bits parallel data as follows:
  - Set CS high then low to allow for data transfer
  - Repeat pulsing the clock line high then low for 9 cycles to store data in shift register



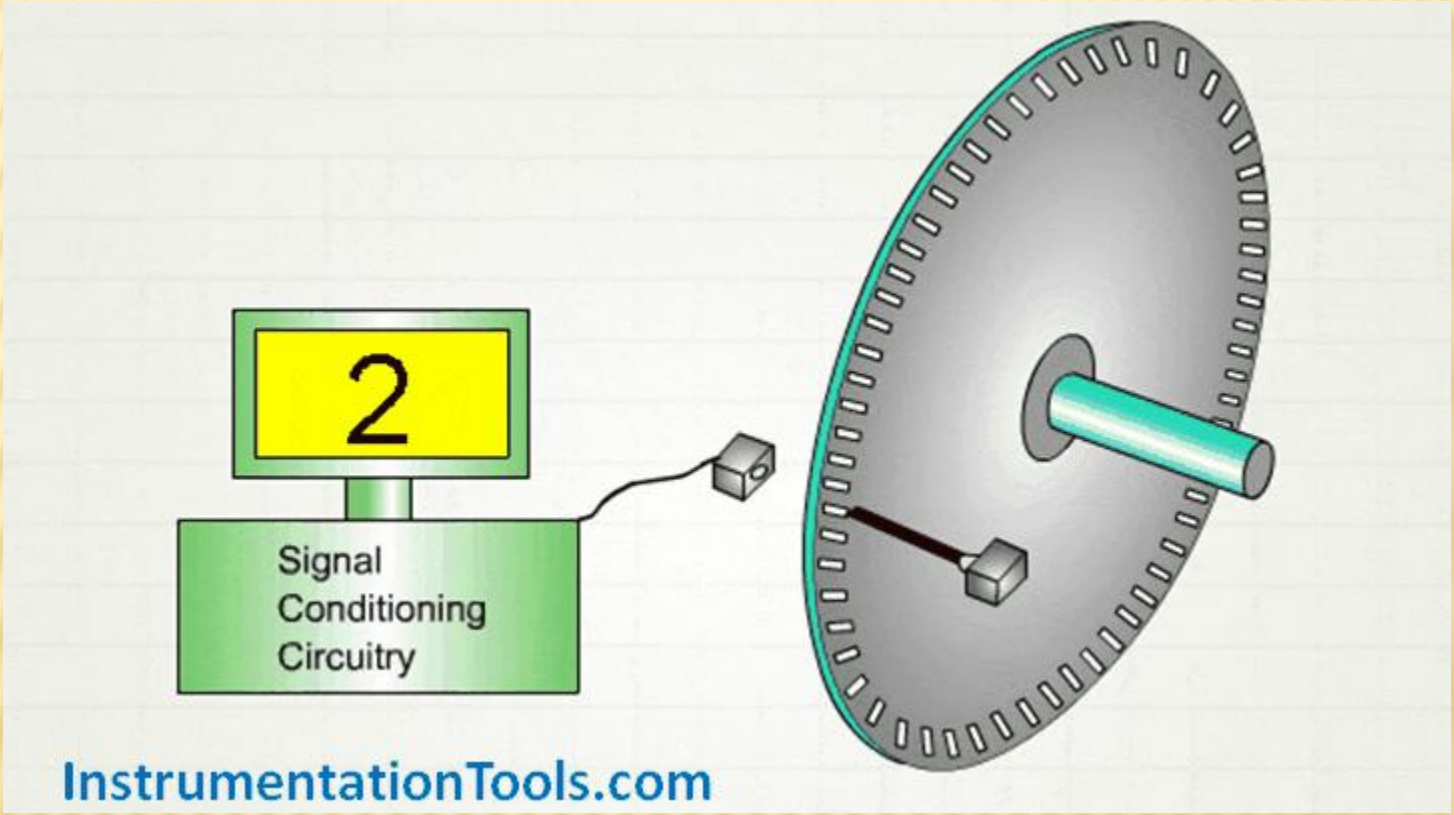
SHAFT ENCODER

INCREMENTAL ENCODER

- Shaft encoders are required as a fundamental feedback sensor for motor control (Chapters 4 and 5).
- The most widely used encoder techniques are either magnetic encoders or optical encoders
- Standard optical encoders use a sector disk with black and white segments together with an LED and a photo-diode.
- The photo-diode detects reflected light during a white segment, but not during a black segment.
- If this disk has 16 white and 16 black segments, the sensor will receive 16 pulses during one revolution.
- Magnetic enco tating disk on the motor shaft w 16) mounted in a circle.
- Every revoluti nets past the Hall sensor and the ne encoder line.



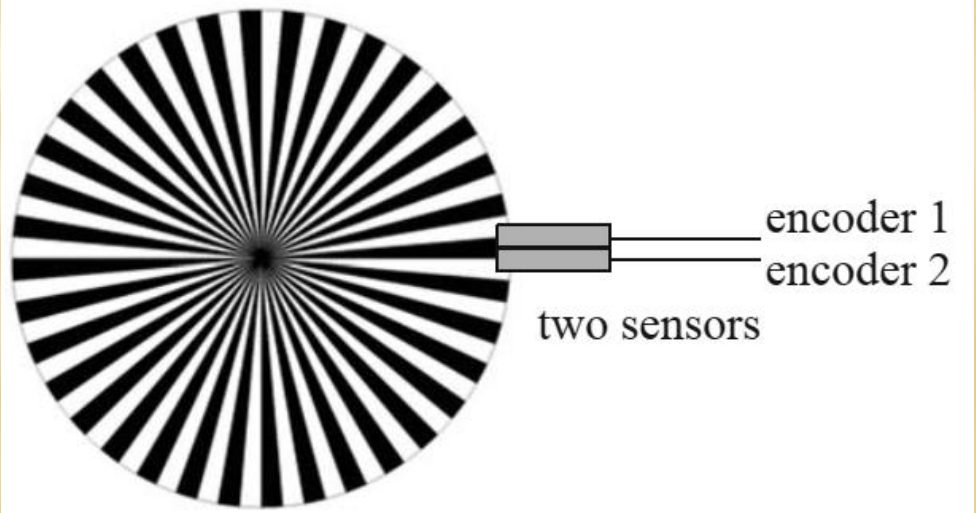
INCREMENTAL ENCODER



Comparison	Optical Encoder	Magnetic Encoder
Working Principle	Based on **light reflection** from a rotating disk divided into black and white segments, using an **LED** and a **photo-diode**.	Based on **magnetic field detection** using a **Hall-effect sensor** and a rotating disk with **magnets**.
Main Components	Sector disk (black and white), light source (LED), photo-diode sensor.	Rotating disk with magnets, Hall-effect sensor.
Signal Output	Generates pulses when the white areas reflect light to the photo-diode.	Generates pulses when magnets pass by the Hall sensor.
Pulses per Revolution	Equal to the number of white segments (e.g., 16 pulses per rotation).	Equal to the number of magnets (e.g., 16 pulses per rotation).
Accuracy	Very high — suitable for precise position and speed measurement.	Moderate — less accurate but sufficient for general control applications.
Resistance to Dust & Vibration	Sensitive to dust and dirt (light path can be blocked).	More robust and resistant to dust, vibration, and harsh environments.
Typical Applications	Precision robotics, CNC machines, laboratory instruments.	Industrial motors, heavy machinery, automotive systems.
Cost	Usually more expensive due to high precision and optical components.	Generally cheaper and more durable.

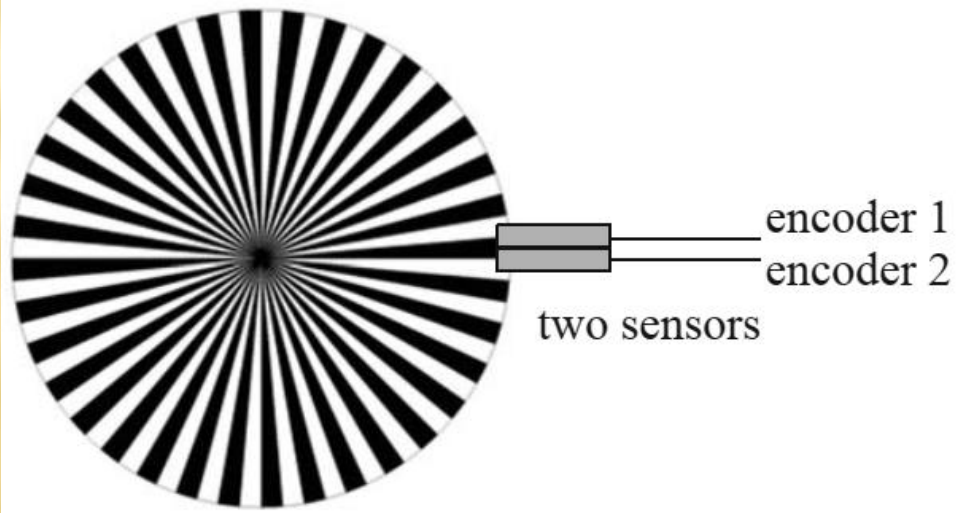
INCREMENTAL ENCODER

- Encoder with only a single magnetic or optical sensor element can only count the number of segments passing by.
- But it cannot distinguish whether the motor shaft is moving clockwise or counterclockwise.
- For this reason most encoders are equipped with two sensors that are positioned with a small phase shift to each other.
- Now it is possible to determine the rotation direction by detecting which of the two sensors first receives the pulse for a new segment.
- If in Enc1 receives the signal first, then the motion is clockwise; if Enc2 receives the signal first, then the motion is counter-clockwise



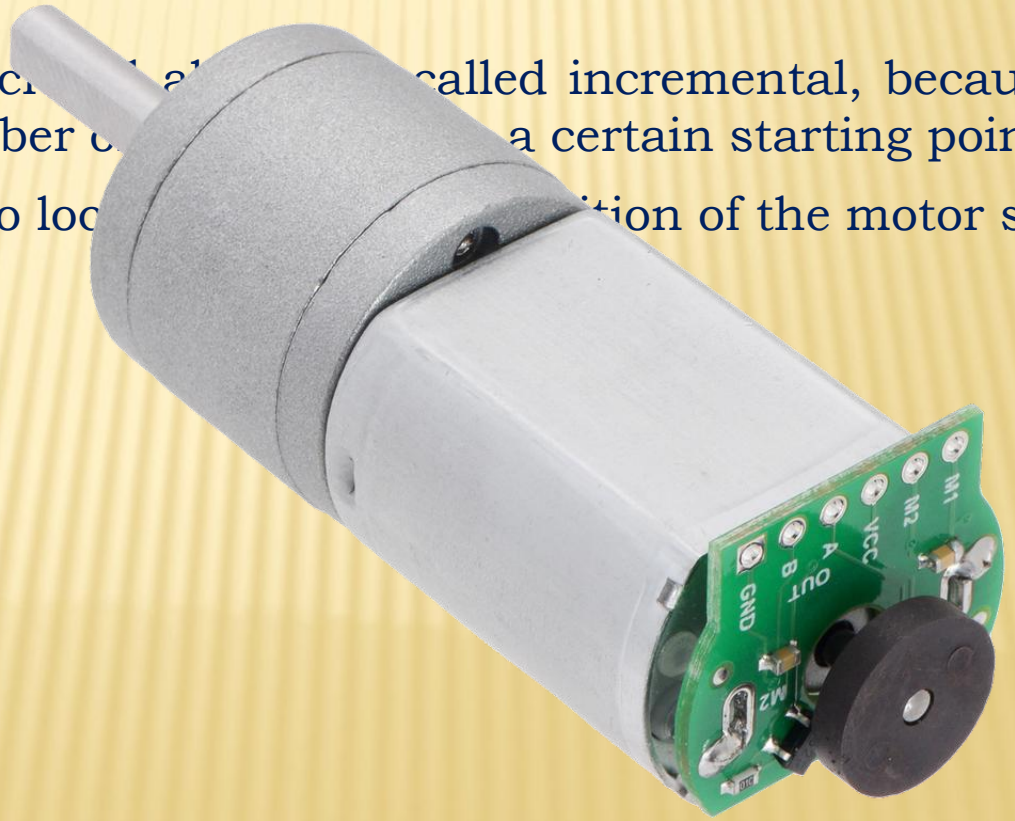
PRACTICAL CONSIDERATION

- Since each of the two sensors of an encoder is just a binary digital sensor, we could interface them to a microcontroller by using two digital inputs.
- However, this would not be very efficient, since then the controller would have to constantly poll the sensor data lines in order to record any changes and update the sector count
- Luckily this is not necessary, since most modern have special input hardware for cases like this.
- They are usually called “pulse counting registers” and can count incoming pulses up to a certain frequency completely independently of the CPU.
- This means the CPU is not being slowed down and is therefore free to work on higher-level application programs.



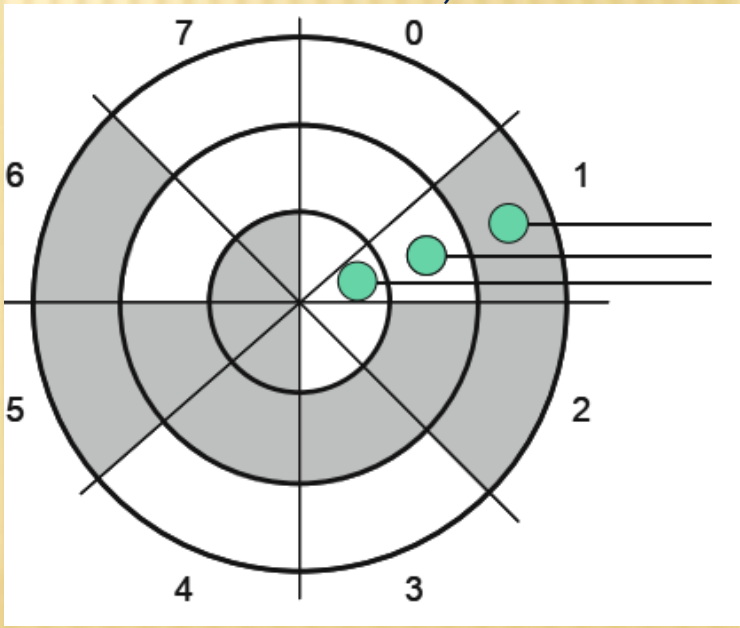
SHAFT ENCODER MOUNTING

- Encoders are usually mounted directly on the motor shaft (that is before the gear box), so they have the full resolution.
- For example, if we have an encoder which detects 16 ticks per revolution and a gearbox with a ratio of 100:1 between the motor and the vehicle's wheel, then this gives us an encoder resolution of 1,600 ticks per wheel revolution.
- Both encoder types desc… a… …called incremental, because they can only count the number o… …a certain starting point.
- They are not sufficient to loc… …ition of the motor shaft.



ABSOLUTE SHAFT ENCODER

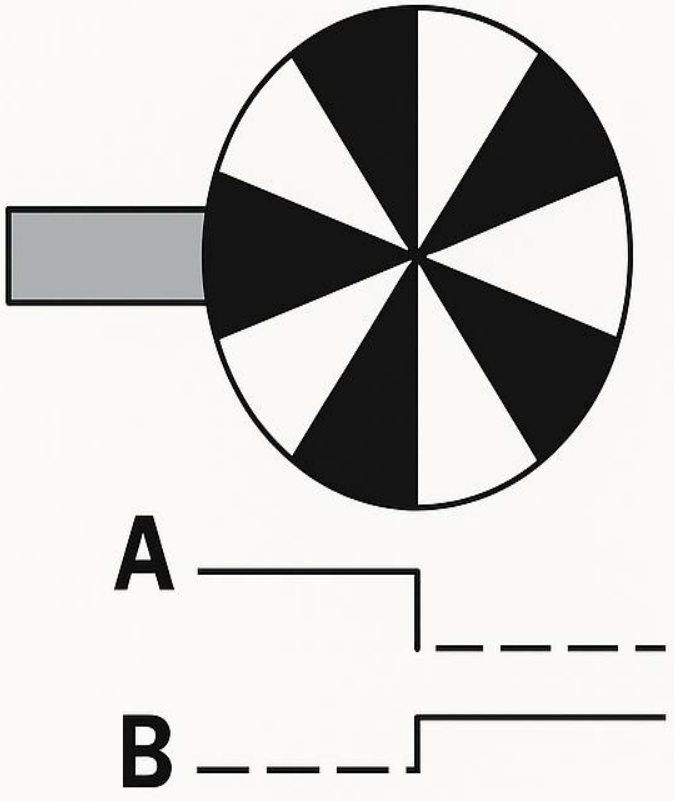
- Absolute encoders measure absolute or true angular position.
- This contrasts with incremental encoders which measure the change in angular position.
- Usually it consists of a Gray-code disk with a set of sensors.
- The number of sensors determines the resolution of this encoder (in the example there are 3 sensors, giving a resolution of $2^3 = 8$ sectors).



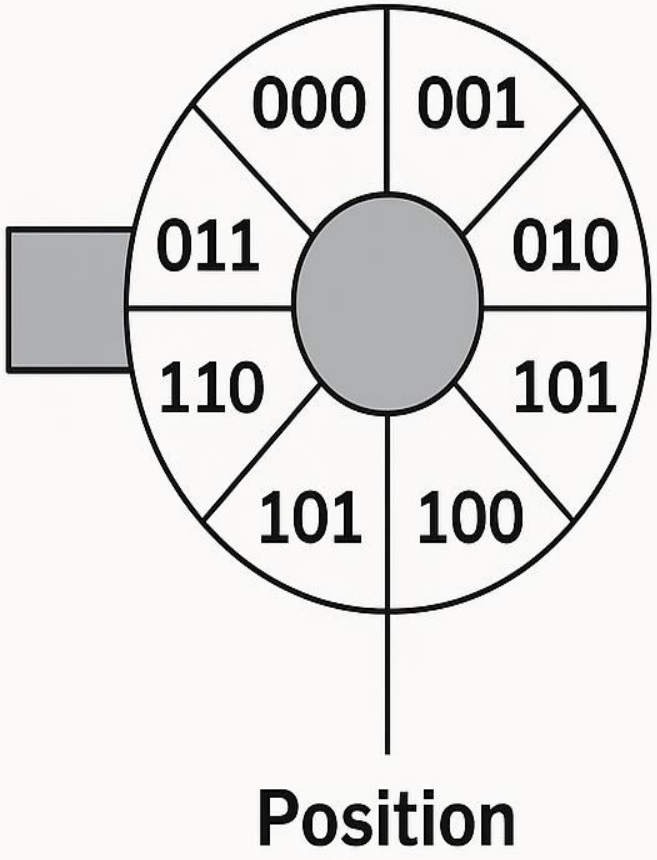
- Note that for any transition between two neighboring sectors of the Gray code disk only a single bit changes.
- This would not be the case for a standard binary encoding
- This is an essential feature of this encoder type, because it will still give a proper reading if the disk just passes between two segments.

For binary encoding the result would be arbitrary when passing between 111 and 000

INCREMENTAL ENCODER



ABSOLUTE ENCODER



🔹 Comparison Between Incremental and Absolute Encoders

Feature	Incremental Encoder	Absolute Encoder
Measurement Type	Measures the *change* in angular position	Measures the *true* or *absolute* angular position
Resolution	Limited by number of pulses per revolution	Determined by number of sensors (bits), very high precision
Power Loss Behavior	Loses position when power is off	Retains position even after power loss
Output Signal	Series of pulses (A, B, sometimes Z)	Digital binary or Gray code representing position
Reference Requirement	Needs a reference or zero point to start counting	No reference needed — each position has a unique code
Main Advantage	Simple, low cost, easy to implement	High accuracy, reliable position reading
Typical Use	Speed measurement, motor feedback	Position control, robotics, automation systems

A/D CONVERTER

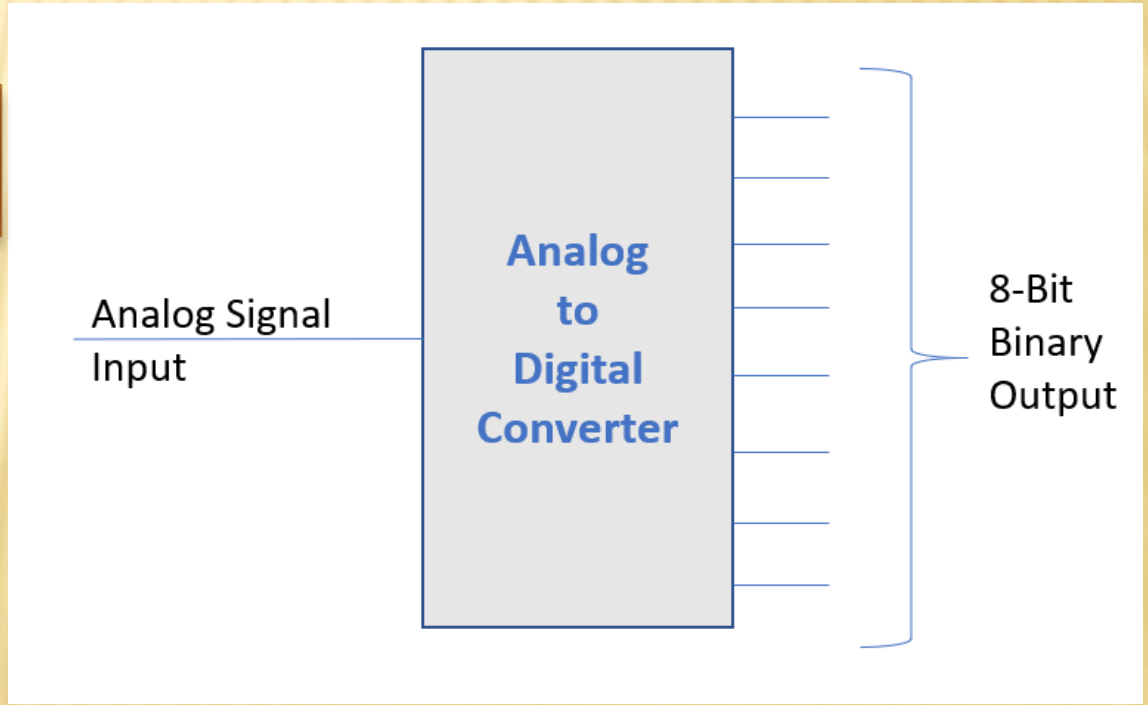
ANALOG TO DIGITAL CONVERTER A/D

- An A/D converter translates an analog signal into a digital value.
- The characteristics of an A/D converter include
  - Accuracy: the number of bits per value (for example 10-bits A/D converter)
  - Speed: maximum conversion rate (for example 500 conversions/second).
  - Measurement range: expressed in volts (for example $0 \rightarrow 5V$).
- The converted digital value can be calculated by the following equation

$$Digital = integer\left(V_{in} \times \frac{2^n}{V_{REF}}\right)$$

$n = number\ of\ bits$

$V_{REF} = reference\ voltage$



A/D OUTPUT

- **Parallel interface**: for example up to 8 bits of accuracy.
- **Synchronous Serial interface**
  - It does not impose any limitations on the number of bits per measurement.
- Many A/D converter modules include a multiplexer which allows the connection of several sensors

